

AI and Policy Memos for International Conflict and Negotiation

INTL 3400

Professor Diamanti-Karanou

Fall 2026

Discussion

- Has anyone ever used ChatGPT or another generative AI?
- What do you think it did well or poorly?
- Were there any aspects of the responses you found odd or misleading?

Workshop Agenda

- Generative AI
 - AI and 'Truth'- Confabulation/Hallucination
- Using ChatGPT and Claude to inform your Policy Memos (Demonstration and Exercise)
- Ethics in AI
 - Bias in AI
 - Countering Bias
 - Personal Impacts
- AI-Powered Literature Review Tools
- Conclusion

Slides, handouts, and data available at <https://bit.ly/3RIOUnb>

Generative AI

*Feel free to ask questions at any point
during the presentation!*

Important questions

- How do human biases impact generative AI model outputs?
- How can we counter the weaknesses of current AI models?
- How can we integrate generative AI with other tools and practices?

Vocabulary

- Artificial Intelligence (AI): A technology that “learns” from datasets to solve problems and mimic human intelligence.
- Large Language Model (LLM): Powerful AI systems designed to understand, generate, and manipulate human language. E.g. GPT4
- Generative AI: It uses AI algorithms that are trained on big datasets in order to create new content, including text, images and audio, in response to a query or prompt. E.g. ChatGPT, Claude, DALL·E, Midjourney.
- Bias in AI: The biased outputs due to human biases that existed in original training data or skew the AI algorithm. Could result in potential harms.

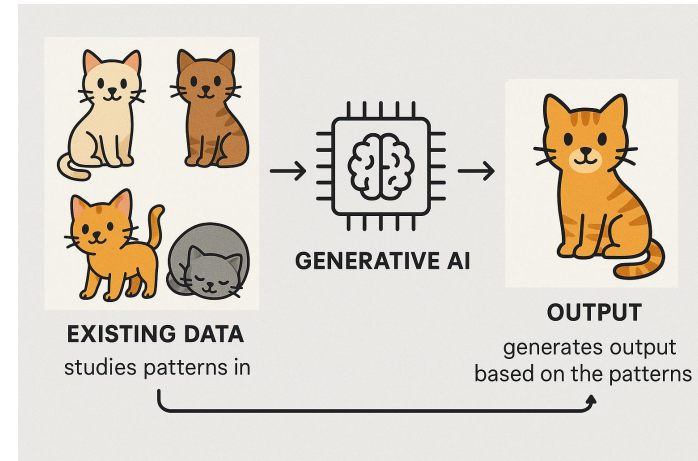
Generative AI: Introduction

AI's like ChatGPT and Claude Sonnet 5 (which generate text) and DALL-E and Midjourney (which generate images) work by training on very large datasets of information and then mimicking that training data to produce “new” outputs.

Generative AI: How it Works

- Generative AIs are fundamentally based on statistics. A text-generation AI like ChatGPT generates words according to patterns learned from the training data.
- Similarly, an image-generation AI learns patterns from existing data and generates images that “looks right” but do not exist in the data. E.g. how AI generates an image of a cat.

Generated by GPT4o
Aug 15th 2025



Example Prompt: Claude 4.6 (1/2)

Prompt: *In a few sentences, explain the history of the UN Department of Political and Peacebuilding Affairs.* (Claude Sonnet 4.6, Accessed June 15th, 2026)

The UN Department of Political and Peacebuilding Affairs (DPPA) was established in 1992 as a department of the UN Secretariat, headquartered in New York City. It plays a central role in UN efforts to prevent deadly conflict and build sustainable peace, monitoring and assessing global political developments to detect potential crises and devise effective responses, while also providing support to the Secretary-General and his envoys in peace initiatives and to UN political missions around the world.

(Continued on following slide)

Example Prompt: Claude 4.6 (2/2)

Prompt: *In a few sentences, explain the history of the UN Department of Political and Peacebuilding Affairs.*

Response: **(From previous slide)** Over the years, the department has expanded its professional capacities in conflict mediation and preventive diplomacy, managing field-based political missions across Africa, Central Asia, and the Middle East, and also overseeing UN electoral assistance to member states.

What questions might you still have after reading this overview?

How the United Nations Paragraph Was Generated

- A large language model analyzed writing on the United Nations and identified common patterns
- These patterns are recreated in the generated text
 - Using keywords related to the United Nations (member states, charter, peacekeeping etc.) and a tone common to academic writing
 - Arrives at history of the United Nations by combining commonly repeated ideas

Exercise: Prompting Claude

Try asking Claude to explain in a few sentences a topic you are familiar with and critically consider the response:

- What did it get right?
- What did it get wrong?
 - Why do you think it made those mistakes?
- Did it miss anything important?

AI And 'Truth'

*Feel free to ask questions at any point
during the presentation!*

Confabulation/Hallucination

- Confabulation refers to the phenomenon where large language models (LLMs) generate incorrect, nonsensical, or fabricated information, even when presented with seemingly clear and accurate prompts.
- AI confabulation is also referred to as hallucination.

Generative AI and “Truth”

- Text-generation AIs aim to produce text that is grammatically correct and linguistically probable.
 - They do not understand “facts,” only patterns of word use.
- They can generate truthful text, but also frequently create confabulations/hallucinations.
 - When asked to generate citations, they can generate plausible-looking but fake sources.
 - They may link real but irrelevant sites as sources for made-up facts. They may also invent URLs that do not work and have never worked.

Truth and AI: Example Citation List (1/2)

Prompt: “Generate 4-5 academic citations on human rights and labor in Vietnam.” (Claude Sonnet 5, August 3, 2026)

1. [Dang, M. H. \(2024\)](#). Applying feminist legal principles to achieve gender equality in Vietnam's labor legislation. *Cogent Social Sciences*, 10(1), Article 2350566.
2. [\(2021\)](#). Child labor, human rights and social justice in Viet Nam. *Peace Review: A Journal of Social Justice*, 33(3).

Correct.

Missing authors.

Truth and AI: Example Citation List (2/2)

3. [Nguyen, \[Author\]](#). (2021). Human rights in Vietnam – A current reality. *The Russian Journal of Vietnamese Studies*, 5(4).

Missing author information.

4. [\(2024\)](#). Labor regimes, global production networks, and state–society relations: Assessing the impact of the EU–Vietnam Free Trade Agreement on labor in Vietnam. *Economic Geography*.

Missing author and journal edition information.

Questions:

From these citations, what do you think these articles have in common?

News: Lawyer Cited AI Fake Cases (1/2)

- In 2026, [CBS News reported](#) that a senior lawyer in an Australian court had to apologize “for filing submissions in a murder case that included fake quotes and nonexistent case judgments generated by artificial intelligence.” (CBS, 2026)

World

**Lawyer apologizes for fake quotes,
fabricated judgments generated by AI in
murder case**

Updated on: January 27, 2026 / 6:21 AM EST / CBS/AP

 Add CBS News on Google

News: Lawyer Cited AI Fake Cases (2/2)

- The fake submissions included fabricated quotes from a speech to the state legislature and nonexistent case citations purportedly from the Supreme Court. (CBS, 2026)
- "The ability of the court to rely upon the accuracy of submissions made by counsel is fundamental to the due administration of justice," Elliott (the judge who received the hallucinated files) added.

[World](#)

**Lawyer apologizes for fake quotes,
fabricated judgments generated by AI in
murder case**

Updated on: January 27, 2026 / 6:21 AM EST / CBS/AP

 Add CBS News on Google

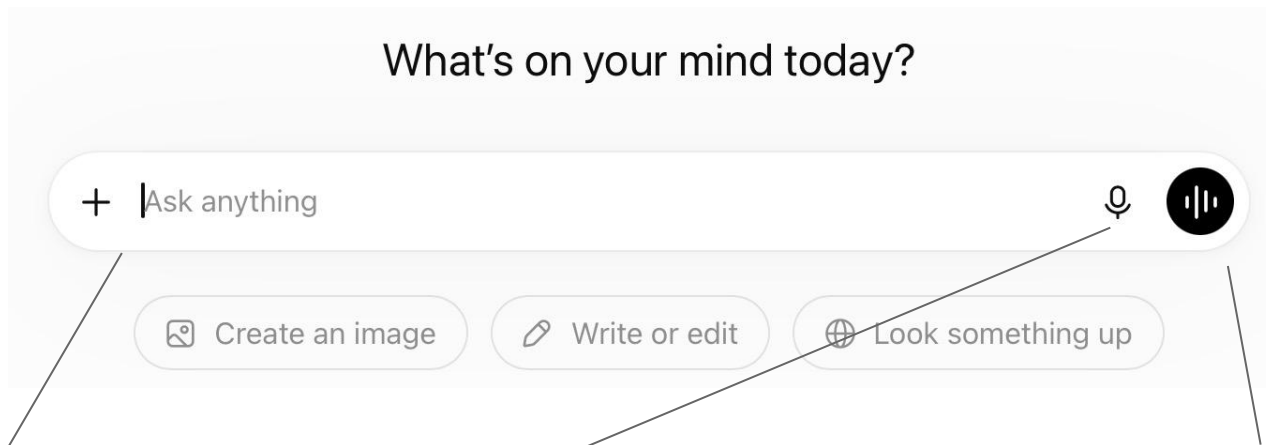
Truth: Generative AI summary of literature

- AI may inaccurately attribute concepts to individuals or omit co-contributors and earlier influences.
- It may lack citations or invent fake references.
- It may provide a surface level summary that is over simplified and misleading.
- It may overemphasize sources it has access to, such as those that are not in a paywalled database, and which are often older.

Using ChatGPT and/or Claude for Policy Memo Research

*Feel free to ask questions at any point
during the presentation!*

Prompting ChatGPT (1/2)



Enter prompt
in text format

Enter prompt as speech
(transcribed automatically)

Use voice mode

Prompting ChatGPT (2/2)

Upload files

📎 Upload photos & files

📁 Recent files >

🖼️ Create image

💡 Thinking

🔍 Deep research

🌐 Web search

⋮ More >

📁 Projects >

Other tools

Follow publisher policies if uploading copyright material to an AI system. See [Northeastern recommendations](#) for more details.

ChatGPT Results

Can you tell me about the psychology of cats?

Initial prompt

Absolutely! The psychology of cats is fascinating because it combines their evolutionary history as solitary hunters with their ability to form social bonds with humans and other animals. Here are some key aspects:

1. Evolutionary roots:

Cats evolved as solitary hunters, which shapes much of their behavior. Unlike dogs (descended from pack animals), cats are more independent, territorial, and selective about social interactions.

Response

2. Social behavior:

Although often labeled "aloof," domestic cats can form deep attachments. They recognize their humans' voices, show preference for familiar people, and may follow routines around food and play. Their affection often comes in subtle ways: slow blinking, rubbing against you, or sitting nearby.

3. Communication:

Cats communicate with body language, scent marking, and vocalizations. Interestingly,

Enter subsequent prompts as text or speech

Use voice mode

Ask anything

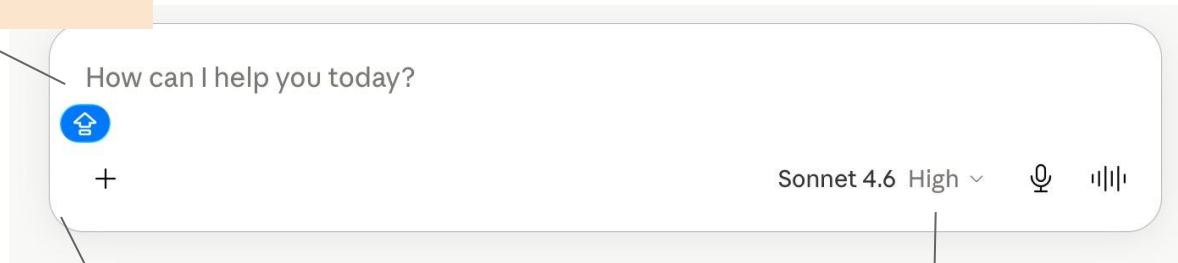
+ Tools

ChatGPT can make mistakes. Check important info.

Prompting Claude

Enter prompt
in text format

Follow publisher policies if uploading
copyright material to an AI system.
See [Northeastern recommendations](#)
for more details.



Upload files

Select models

Prompting Claude (1/2)

The image shows a screenshot of the Claude web interface. A settings menu is open, displaying options like 'Use style', 'Extended thinking' (with a toggle switch), 'Web search' (with a toggle switch), 'Add connectors', and 'Manage connectors'. A red box highlights the 'Settings' icon in the bottom left of the chat input area. A line points from this icon to the 'Open menu' text in an orange box below. Another line points from the 'Extended thinking' toggle switch to a text box that says 'Allow Claude to think out loud for longer for trickier questions and challenges.' To the right, a list of styles is shown: Normal, Learning, Concise, Explanatory, Formal, and a '+ Create & edit styles' option. The model name 'Claude Sonnet 4' is visible at the bottom right of the chat area.

Use style

Extended thinking

Web search

+ Add connectors

Manage connectors

Search menu

Reply

+ [Settings icon] Research

Open menu

Normal

Learning

Concise

Explanatory

Formal

+ Create & edit styles

Allow Claude to think out loud for longer for trickier questions and challenges.

Claude Sonnet 4

Prompting Claude (2/2)

SM Can you provide a list of common biases in AI generated literature reviews?

Here are common biases that can occur in AI-generated literature reviews:

Selection and Search Biases

- **Training data bias** - AI models reflect the biases present in their training datasets, which may overrepresent certain types of research, languages, or publication venues
- **Recency bias** - Overemphasis on recent publications while undervaluing foundational or older work that remains relevant
- **Publication bias** - Tendency to focus on published studies while missing important unpublished research, gray literature, or negative results
- **Language bias** - Disproportionate emphasis on English-language publications, missing valuable research in other languages

Analytical and Interpretive Biases

- **Confirmation bias** - Selectively highlighting studies that support predetermined conclusions while downplaying contradictory evidence
- **Cherry-picking** - Focusing on specific findings from studies while ignoring conflicting results within the same research

Reply to Claude...

+ ⓘ 🔍 Research

Claude Sonnet 4 ▾



Initial prompt

Initial response

Enter subsequent prompts as text

Demonstration Overview:

- Sample Policy Memo topic: You are an advisor to Armenian Prime Minister Nikol Pashinyan. How should he move forward in negotiations with his Azerbaijani counterpart over peace in the Nagorno-Karabakh region?
- Questions likely to be useful for this policy memo: What is the historical significance of the Nagorno-Karabakh region to Armenia and Azerbaijan? How did the fall of the Soviet Union affect the dispute? What is the current status of the Nagorno-Karabakh region and what recent history has determined that status? What are the human costs of the ongoing dispute? What mechanisms are at the disposal of Prime Minister for resolving the conflict? If similar conflicts have been resolved successfully, how did that happen?

Demonstration (1/4):

Question: “What is the historical significance of the Nagorno-Karabakh region to Armenia and Azerbaijan? How did the fall of the Soviet Union affect the dispute? Provide at least 3 APA-style citations for your summary.” (Claude Sonnet 5, August 9, 2026)

[View Response](#)

As you look through the response consider: How and where would you go to check the accuracy of this response (try exploring [Northeastern library databases](#))? Which of the AI provided sources might be most useful? Once you have verified the accuracy of the information, how would you describe the dispute from an Armenian perspective?

Demonstration (2/4):

Question: “What is the current status of the Nagorno-Karabakh region and what recent history has determined that status? What are the human costs of the ongoing dispute? Provide at least 3 APA-style citations for your summary.” (Claude Sonnet 5, August 9, 2026)

[View Response](#)

As you look through the response consider: How accurate is the summary (try comparing what is here to reliable news sources)? What does the current status of the conflict suggest about Armenia’s options for the future? How might you think about the human cost of the conflict as you write your memo? Which of the citations provided by Claude might be most useful for further research?

Demonstration (3/4):

Question: “How can the Prime Minister of Armenia promote peace between Armenia and Azerbaijan over the Nagorno-Karabakh region while also protecting the interests of Armenia? Provide at least 3 APA-style citations for your summary.” (Claude Sonnet 5, August 9, 2026)

[View Response](#)

As you look through the response consider: Is anything missing or inaccurate from this response (news sources or the [Northeastern library](#) might be useful here)? Which of the AI-suggested sources might be most useful for further reading to inform your policy memo? As an advisor to Armenia, how would you interpret this summary? Where might you agree or disagree, and how might that inform your memo?

Demonstration (4/4):

Question: “How have conflicts similar to the one over Nagorno-Karabakh been successfully resolved? Provide at least 3 APA-style citations for your summary.”
(Claude Sonnet 5, August 13, 2026)

[View Response](#)

As you look through the response consider: Is anything missing or inaccurate from this response (the [Northeastern library](#) resources might be useful here)? Which of the AI-suggested sources might be most useful for further reading to inform your policy memo? Do you see any relevant lessons from the comparisons made in the response? If so, how might you apply them to this case? Do any of the drawn-out conflicts mentioned offer warnings about options you had been considering?

Your turn

- Design questions with which to prompt Claude or ChatGPT on the history/current status/human rights implications/drawbacks and benefits of different approaches for one of the conflicts you plan to write a memo about (from the perspective of the country you serve as an advisor for). For better results, ask several separate questions rather than one multi-part question. Make sure to also request academic citations.
- Analyze the responses the same way we did for the Azerbaijan/Armenia example. What information might be most relevant for your memo (be sure to check its validity)? Which sources provided by the AI may be most useful? Finally, consider next step and how to incorporate what you have found today into your memo's approach to the conflict.

Ethics in AI

*Feel free to ask questions at any point
during the presentation!*

Ethics: Generative AI “Originality”

- Some argue that all AI-generated output constitutes plagiarism and copyright infringement, since it is remixing training data that was scraped from the internet without permission from the original creators.
- Many AI companies are [facing lawsuits](#) from people whose content was used as training data without their consent.
- Some publication venues, like the *Science* journals, have made it an [official policy](#) that AI does not meet the standard for authorship and require authors to disclose use of AI.

Ethics: Generative AI Training

- AI training data is sometimes supplemented by labels (annotations) added by people ([Amironesei and Díaz, 2024](#)). These labels can worsen bias in training datasets.
- People from middle and low income countries often labor in poor working conditions to annotate data for clients in high income countries. Fieldwork by [Muldoon et al.](#) (2023) revealed that workers faced traumatizing content, in addition to experiencing discrimination in the workplace and receiving low wages.
- Public awareness can help pressure companies to adopt good practices. For more information on fair labor in AI, see the report [AI for Fair Work: From principles to practices](#) by [Fairwork](#)

Ethics: Environment (1/2)

- Training and using AI requires processing very large amounts of data, which is done in data centers
- These data centers can have a negative impact on the environment and communities
- Given their intensive energy and water demands, data centers can [worsen local water scarcity](#) and [increase electricity prices](#)
- To explore how the energy use of AI compares to other digital tasks, check out [Jon Ippolito's "What Uses More" app](#)
- For more information on carbon emissions from running computer code (including for AI) check out the Python library [CodeCarbon](#).

Ethics: Environment (2/2)

- A large data center may use 5 million gallons of water per day, equivalent to the water use of a town of about 10,000 to 50,000 people.
- One 100-word AI prompt uses roughly the equivalent of one bottle of water.
- How do data centers use water? It is calculated from water used in the centers themselves (mostly for cooling), water used in power plants that supply energy to data centers, and water consumption in the manufacturing of processor chips.

Source: ['Data Centers and Water Consumption' by Miguel Yañez-Barnuevo \(EESI\)](#)

One Note on Ethics and AI

- These slides (and the ones which follow) do not cover all of the ethical considerations of AI development/use. This is a quickly changing field, and there are no complete guidelines or perfect authorities on ethics and AI. Much remains to explore beyond what we are discussing today.
- Starting points for further exploration:
 - [Ethics in AI: Why It Matters](#)
 - [Recommendation on the Ethics of Artificial Intelligence](#)
 - [A Guide to AI Ethics](#)

Bias in AI

*Feel free to ask questions at any point
during the presentation!*

What is AI bias?

- The biased outputs due to human biases that exist in the original training data or skew the AI algorithm.
 - Could result in potential harms.
- For example, when AI was used to summarize medical notes, “Google’s AI tool Gemma described men’s health issues with terms like “disabled,” “unable,” and “complex” significantly more often than women’s, who were often framed as more independent despite similar needs, an alarming gender bias trend.” (Source: [crescendo.ai](https://www.crescendo.ai))
- What is the ideal “unbiased” scenario? Is an “unbiased” scenario possible?

Bias in AI: Summary

- AI training data reflects the injustices and biases of the society that produced it.
- These biases can be amplified when they are input as training data into an AI, because they seem to be the “right” answers ([Dwivedi et al. 2023](#)).

Bias in AI: Study

A study from Cornell University tested seven LLMs on scenarios that International Relations practitioners might present them with (on questions of military escalation, military and humanitarian intervention, cooperative dynamics, and alliance dynamics).

The results:

- Variation among the models, with some recommending significantly more escalatory actions,
- All models tended to recommend greater escalation when questioned by the 'US' or the 'UK' rather than 'China' or 'Russia.'

(Claude, Sonnet
4.6, 6/17/26)

Source: [Measuring Diplomatic Preferences in LLMs](#)

Bias in AI: Example Citation List 1/2

Example: “Generate a list of 4 academic citations on the origins of international law” AI-Generated Answer:

1. [Koskenniemi, M.](#) (2001). *The Gentle Civilizer of Nations: The Rise and Fall of International Law, 1870–1960*. Cambridge University Press.
2. Grotius, H. (1625). [Mare Liberum](#) and [De Jure Belli ac Pacis](#) [On the Law of War and Peace]. (F. W. Kelsey, Trans., 1925). Carnegie Endowment for International Peace.

Question: what perspectives might be missing?

(Claude, Sonnet 4.6, 6/17/26)

Feel free to ask questions at any point during the presentation!

Bias in AI: Example Citation List 2/2

Example: “Generate a list of 4 academic citations on the origins of international law” AI-Generated Answer:

3. Nussbaum, A. (1954). [A Concise History of the Law of Nations](#) (rev. ed.). Macmillan.
4. Vattel, E. de. (1758). [Le Droit des gens, ou Principes de la loi naturelle](#) [The Law of Nations]. (C. G. Fenwick, Trans., 1916). Carnegie Endowment for International Peace.

Question: what perspectives might be missing?

(Claude, Sonnet 4.6, 6/17/26)

Bias in AI: Types (1/2)

Bias Type	How It Shows Up in the Example	Why It Happens
Western bias	All the recent articles come from scholars based in Western academic institutions, and the historical documents were also produced in Europe.	Training data over-represents English-language, high-impact Western journals.
Prestige bias	Classic, highly cited papers dominate; recent community-based or non-English studies are missing. “Rich get richer.”	The model “learns” that frequently quoted articles are more “salient.”

Bias in AI: Types (2/2)

Bias Type	How It Shows Up in the Example	Why It Happens
Gender / editor influence	All of cited first authors are male despite contributions by female scholars in international relations scholarship.	Historical citation patterns in the corpus perpetuate disparities.

Bias in Results: Claude 4.6 Sonnet

Prompt: *Write a sentence about important moments in public diplomacy in the 20th century.* (Claude Sonnet 4.6, June 17, 2026)

Response: Public diplomacy in the twentieth century was shaped by a series of pivotal moments, including Woodrow Wilson's Fourteen Points address of 1918, which reframed war aims in terms of self-determination and open diplomacy, the establishment of the United States Information Agency in 1953 to project American values during the Cold War, and the Soviet Union's use of Radio Moscow and cultural exchanges to compete for global influence.

What might be missing from this summary?

Countering Bias

*Feel free to ask questions at any point
during the presentation!*

Countering Bias: Women and Diplomacy

Prompt: *Write a sentence about significant women practitioners of public diplomacy.* (Claude Sonnet 4.6, June 17, 2026)

Response: Among the most significant women practitioners of public diplomacy are Eleanor Roosevelt, who leveraged her role as the first chair of the UN Commission on Human Rights to champion universal rights on the world stage, Jeane Kirkpatrick, who as the first female US Ambassador to the United Nations wielded sharp ideological influence during the Cold War, and Madeleine Albright, whose tenure as US Secretary of State made her a prominent face of American diplomacy abroad at a critical post-Cold War moment.

Countering Bias: Global South Diplomacy

Prompt: *Write a sentences about practitioners of public diplomacy from the Global South.* (Claude, Sonnet 4.6, June 17, 2026)

Response: Practitioners of public diplomacy from the Global South have made profound contributions to shaping international discourse, among them Julius Nyerere of Tanzania, whose advocacy for African socialism and South-South cooperation gave moral weight to postcolonial self-determination and Jawaharlal Nehru of India, whose co-founding of the Non-Aligned Movement in 1961 offered newly independent nations an alternative to Cold War bloc politics.

Countering Bias: Discussion

- What groups of diplomats are still excluded from these summaries?
- What other biases may they contain?
- What are biases that we cannot counter by changing prompts?

Countering Bias: Prompts and Feedback

- Bias can be partly countered through careful prompt design and feedback ([Dwivedi et al.](#), 2023). However, these methods do not remove bias intrinsic in the model ([Shin et al.](#), 2024)
- Practices for countering bias
 - Identify gaps or inconsistencies in generative AI responses.
 - Use additional inquiry and prompt revision to help fill in gaps.
 - Double check responses with other sources.

Personal Impacts

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during the presentation!*

Critical Thinking

- Recent studies show that the impact of AI on critical thinking depends on how it is used.
- In instances where AI is being used to assist in accessing and analyzing information and in self-directed learning it may support critical thinking ([Melisa et al.](#), 2025; [Deep and Chen](#), 2025)
- However, critical reflection and evaluation and writing skills may be hampered by over-reliance on AI ([Melisa et al.](#), 2025; [Deep and Chen](#), 2025)
- High self-confidence also fosters critical thinking, while high confidence in AI may result in less critical thinking ([Lee et al.](#), 2025)

Language and Expression

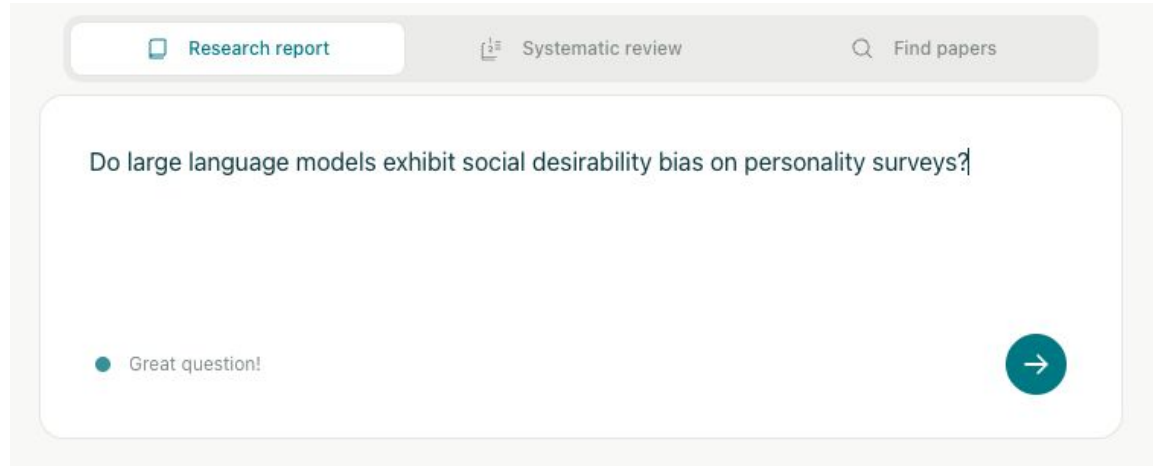
- AI models also vary significantly in their ability to reflect cultural and linguistic nuance depending on the digital prevalence of the language ([Shan et al.](#), 2025).
- While AI improves the efficiency of translation it is not as effective for cultural references ([Charles-Kenechi](#), 2024).
- Use of AI assistance in writing can hamper the expression of cultural nuance and “reinforce Western cultural hegemony” ([Agarwal et al.](#), 2025).

AI-Powered Literature Review Tools

*Feel free to ask questions at any point
during the presentation!*

Elicit

- [Elicit](#) uses AI to generate [literature reports](#)
- Provides feedback on research question



Research question from [Salecha et al.](#), (2024)

Elicit: Results (Research Report)

Social Desirability in Language Models

Research report

Create alertShare

SEPTEMBER 23, 2025

Do large language models exhibit social desirability bias on personality surveys?

Large language models exhibit social desirability bias on personality surveys, with models like GPT-4 and Llama 3 shifting their responses by over one human standard deviation toward more socially favorable traits such as increased extraversion and reduced neuroticism.

ABSTRACT

Large language models exhibit responses that shift toward socially desirable traits when completing personality questionnaires. Salecha et al. (2024a, 2024b) report that models such as GPT-4 and Llama 3 change their survey responses by 1.20 and 0.98 human standard deviations, respectively, toward increased extraversion and reduced neuroticism when the context suggests evaluation. Reverse-coded items reduced but did not eliminate these shifts, ruling out simple acquiescence bias. Lee et al. (2024a) found that adding commitment statements raised Social Desirability Response index scores from a mean of 4.60 to 5.49 while slightly lowering civic engagement scores. Other studies indicate that bias magnitudes are influenced by model sophistication and even synthetic demographic cues, with some assessments (including the Big Five, Short Dark Triad, and HEXACO instruments) revealing systematic trends toward socially preferred responses.

METHODS

We analyzed 10 sources from an initial pool of 50, using 7 screening criteria. Each paper was reviewed for 5 key aspects that mattered most to the research question.

More on methods

Report

Status

Gather sources

50 sources found

Details

Screen sources

10 sources included

Details

Extract data

60 data points extracted

Details

Generate report

Save PDF

Chat

Ask anything about the results

Study	Study Focus	Large Language Model (LLM) Models	Experimental Design	Bias Detection
Salecha et al., 2024a	Social desirability bias in large language models on Big Five personality surveys	GPT-4 (0613), GPT-3.5, Claude 3 Opus/Haiku, PaLM-2, Llama 3 70B Instruct, Llama 2 70B Chat	Systematic variation of question number, context, coding, and temperature; multiple models	Score shifts toward socially desirable traits as number of questions increases; effect sizes in human standard deviations; reverse-coded items; paraphrasing, rewording
Salecha et al., 2024b	Social desirability bias in large language models on Big Five personality surveys	GPT-4 (0613), GPT-3.5, Claude 3 Opus/Haiku, PaLM-2, Llama 3 70B Instruct, Llama 2 70B Chat	Systematic variation of question number, coding, paraphrasing, temperature; multiple models	Score shifts toward socially desirable traits as number of questions increases; effect sizes in human standard deviations; reverse-coded items; paraphrasing, rewording
Lee et al., 2024	Social desirability response (SDR) bias in GPT-4 survey simulations	GPT-4 (gpt-4-1106-preview)	Within-subjects: presence/absence of commitment statement; cross-cultural personas	SDR index (sum of responses), civic engagement index; comparison of commitment statements

Elicit: Results (Find Papers)

Social Desirability Bias in Language Models

Share>Create alert

Q Do large language models exhibit social desirability bias on personality surveys?

Summary of top 4 sources

Copy

Recent research reveals that large language models (LLMs) exhibit significant social desirability bias when completing personality surveys. [Salecha et al. \(2024\)](#) demonstrated that LLMs systematically skew their responses toward socially desirable traits (increased extraversion, decreased neuroticism) when they infer they are being evaluated on personality dimensions. This bias was observed across multiple models including GPT-4, Claude 3, Llama 3, and PaLM-2, with effect sizes reaching 1.20 standard deviations for GPT-4 and 0.98 for Llama 3. The bias persisted despite question randomization and paraphrasing, and reverse-coding reduced but did not eliminate the effect, ruling out simple acquiescence bias. [Lee et al. \(2024\)](#) found mixed evidence for social desirability response bias in GPT-4 when simulating responses from different societies, with commitment statements increasing bias indices but reducing civic engagement scores. [Geng et al. \(2024\)](#) identified fundamental cultural, age, and gender biases in LLM survey responses, emphasizing the importance of analyzing prompt robustness before using LLMs to simulate social surveys.

Sort: Most relevant Filters Export as UPGRADE			
Paper	Abstract summary	Manage Columns	
☐ Large language models display human-like social desirability biases in Big Five personality surveys Aadesh Salecha +5 PNAS Nexus 2024 · 25 citations DOI	Large language models exhibit human-like desirability biases when completing Big Five personality surveys.	Search or create a column Describe what kind of data you want to extract e.g. Limitations, Survival time ADD COLUMNS + Summary + Main findings + Methodology + Intervention + Outcome measured + Limitations Show more	
☐ Large Language Models Show Human-like Social Desirability Biases in Survey Responses Aadesh Salecha +5 arXiv.org 2024 · 12 citations DOI	Large language models exhibit human-like desirability bias when responding to personality surveys.		
☐ Exploring Social Desirability Response Bias in Large Language Models: Evidence from GPT-4 Simulations Sanguk Lee +4 arXiv.org	Large language models like GPT-4 may exhibit social desirability response bias, but the evidence is mixed.		

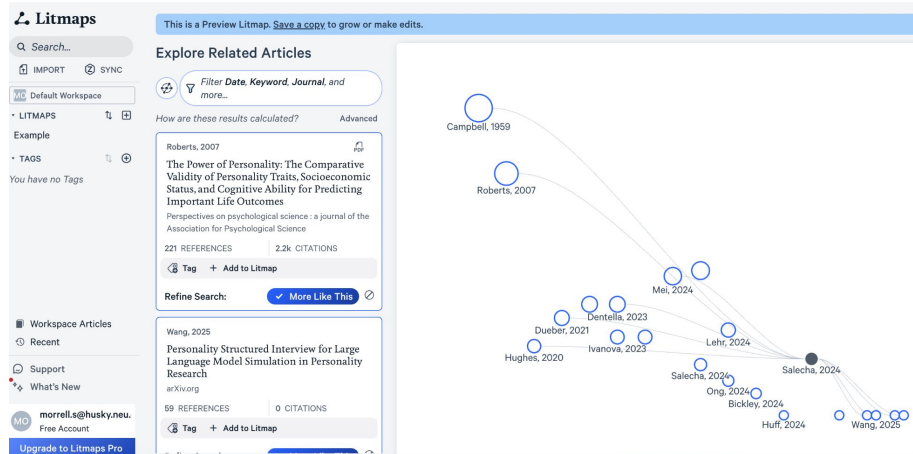
Litmaps

- [Litmaps](#) maps relevant literature based on citations, authors, or textual similarity
- Uses AI to find articles with similar text

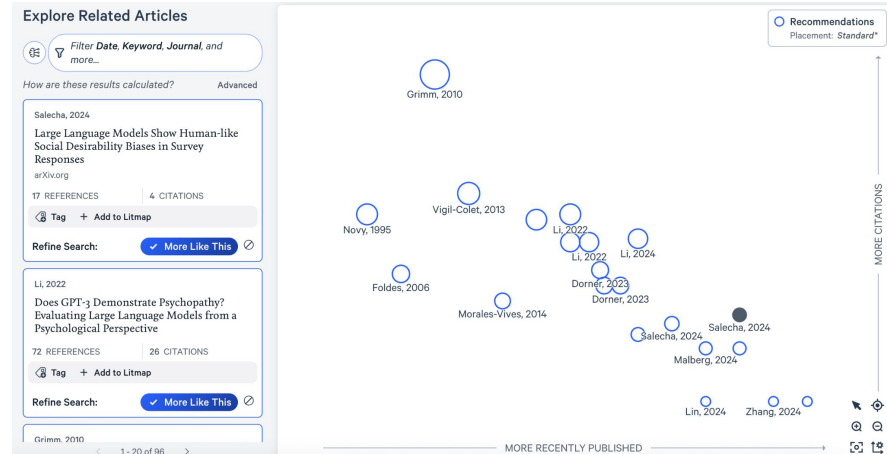
Discover the world of Scientific Literature

🔍 Large language models display human-like social desirability bias.

Litmaps: Results



Mapping by shared references and citations



Mapping by textual similarity

Other tools

- [Scite.ai](#), finding and summarizing papers, provides context for number of citations by explaining how the paper was used in each of its citations (licensed by Northeastern). [Handout here](#).
- [Scopus](#), develops search queries and creates literature summaries in text or image format. (licensed by Northeastern). [Handout here](#).
- [docAnalyzer.ai](#), document analysis.
- [Consensus](#), finds papers and evaluates the extent to which their results agree.
- [Research Rabbit](#), finds and organizes papers.

Conclusion

*Feel free to ask questions at any point
during the presentation!*

Main Points

- AIs, and the data used to train them, are biased
- You can design inputs to help counteract biases
- Always double check AI output
- There are multiple AI tools that may be useful at different points in your work

Reflection

1. Have your perspectives changed on AI after this class? If so, how?
2. How might you use the AI tools differently?

For Further Exploration

DITI Handouts:

[Copyright and fair use handout](#)

[Data Ethics handout](#)

[Data Privacy handout](#)

Northeastern University Resources:

[Northeastern Policy on the Use of AI](#)

[Generative AI in Teaching and Learning](#)

[ChatGPT to Support Reading Development and Critical Thinking](#)

[Standards for the Use of Artificial Intelligence in Research](#)

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NULab Faculty Research on Data, Algorithms, & AI

- [“John Wihbey and Christo Wilson on Tiktok Data Espionage Concerns”](#)
- [“Alan Mislove Co-Authors Research on Discriminatory Ad Algorithms”](#)
- [“John Wihbey Weighs In On AI’s Potential to Impact the 2024 Presidential Election”](#)
- [“Tina Eliassi-Rad Co-Creates New AI Model that Predicts Human Lifespan”](#)
- [“Nabeel Gillani Interviewed by Tech Talk Podcast on AI and Education”](#)

(Cont.) NULab Faculty Research on Data, Algorithms, & AI

- [“John Wihbey Participates in a Panel on Content Moderation”](#)
- [“John Wihbey Comments on Google’s New ‘AI Overview’”](#)
- [“John Wihbey on the Politics of AI”](#)
- [“John Wihbey Interviewed on AI and Epistemic Risk”](#)
- [“Malik Haddad on the Regulation of AI”](#)

How to use AI responsibly EVERY time

- “**Evaluate** the initial output to see if it meets the intended purpose and your needs.”
- “**Verify** facts, figures, quotes, and data using reliable sources to ensure there are no hallucinations [or confabulations] or bias.”
- “**Engage** in every conversation with the GenAI chatbot, providing critical feedback and oversight to improve the AI’s output.”
- “**Revise** the results to reflect your unique needs, style, and/or tone. AI output is a great starting point, but shouldn’t be a final product.”
- “**You** are ultimately responsible for everything you create with AI. Always be **transparent** about if and how you used AI.”
- Source from [AI for Education](#)

AI Plagiarism Checkers

*Feel free to ask questions at any point
during the presentation!*

Plagiarism Checkers: Summary

- Some companies sell tools that claim to identify whether text is AI-generated or human-generated.
- They do this by calculating how statistically predictable each word in the text is.
- AI-checkers can be **wrong**: If a text consistently uses predictable words it is more likely to be labeled as AI generated.

Plagiarism Checkers: Biases

- These tools have the potential for false positives (identifying human texts as AI).
- False positives are especially likely for texts by writers for whom English is not their first language or for writers who have autism, ADHD, dyslexia, or related neurodivergence.
- To make things worse, writers can often reduce their “AI score” by using an AI to reword their essays.

Thank you!

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